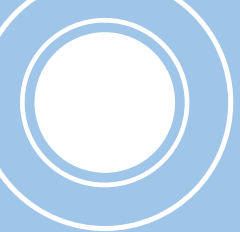




Predicting Controllable Flight Delays Using Machine Learning: A North Carolina Flight Operations Analysis



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01

Background & Business Case



Project Overview & Motivation

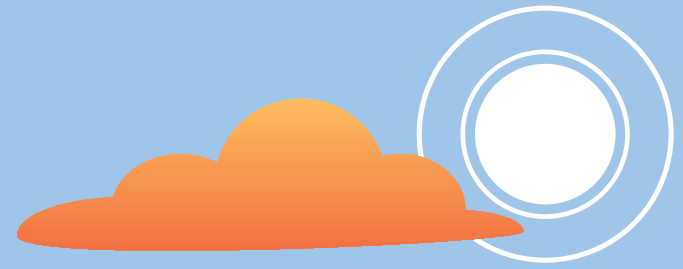


- Aviation industry → complex scheduling and tightly controlled environments
- Flight delays are more than an inconvenience
 - Fuel burn
 - Crew overtime
 - Rebooking fees
 - Passenger discontent
- Flight delays impact brands
 - Passenger itineraries
 - Brand loyalty

The Strategic Need: To minimize inefficiencies, stakeholders must focus on actionable metrics - delays they can actually prevent



Why North Carolina?



- Serves as a major node in the US aviation network
- Home to major players:
 - Charlotte Douglas Int'l (CLT)
 - Raleigh-Durham (RDU)
 - Greensboro (GSO)
- Relevant to us as students at Elon
- Other minor airports:
 - Asheville (AVL)
 - Wilmington (ILM)
 - Jacksonville (OAJ)
 - Fayetteville (FAY)
 - Concord-Padgett (USA)
 - Coastal Carolina (EWN)

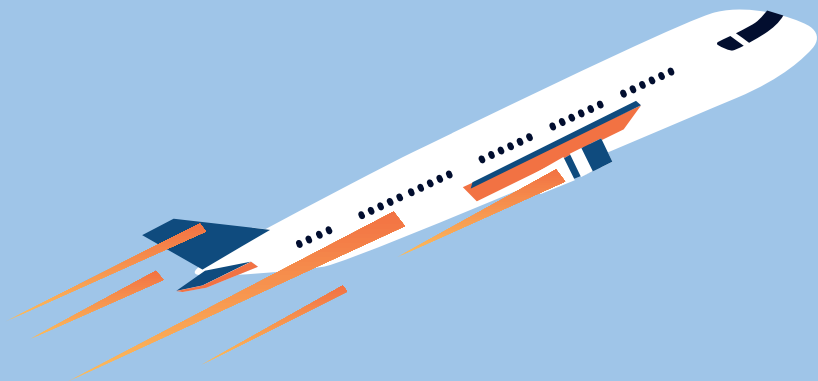


Problem Statement



- **Current state** → airlines and airports often react to controllable delays only after they occur
- **The problem** → the lack of proactive identification for flights at high risk for specific, avoidable disruptions
- **Project goal** → develop predictive algorithms using variables to flag flights with a high probability of a controllable delay
- **Outcome** → resources allocated before the delay happens





02

Data Understanding & Preparation



Dataset Profile

- Sourced from Kaggle
 - 2024 US Bureau of Transportation Statistics
- Focused on domestic flights operating out of North Carolina
- Started with 35 columns, narrowed to 17 relevant predictors
 - Time, date, airtime, location, reason for delay

kaggle



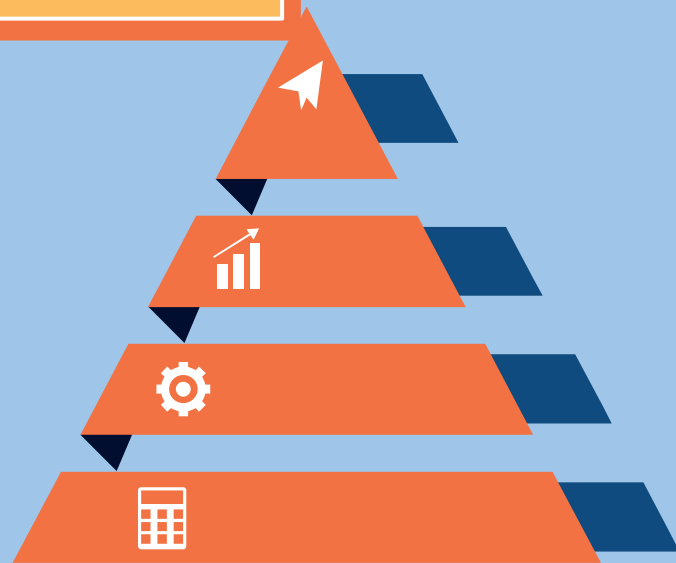
Preprocessing

- Handled missing data by filling zeros
- Column selection by cutting irrelevant categories
- Target creation
 - Controllable delay - carrier delay, NAS congestion, security delay
 - Uncontrollable delay - weather, late-arriving aircraft



Feature Engineering

- Extracted departure hours and binned them
 - Night (0-6), Morning (6-12), Afternoon (12-18), Evening (18-24)
- Created a temporal feature of weekend indicator
- Transformed categorical variables (operating carrier, origin airport) into dummy variables
- Managed balanced and unbalanced data with minority class





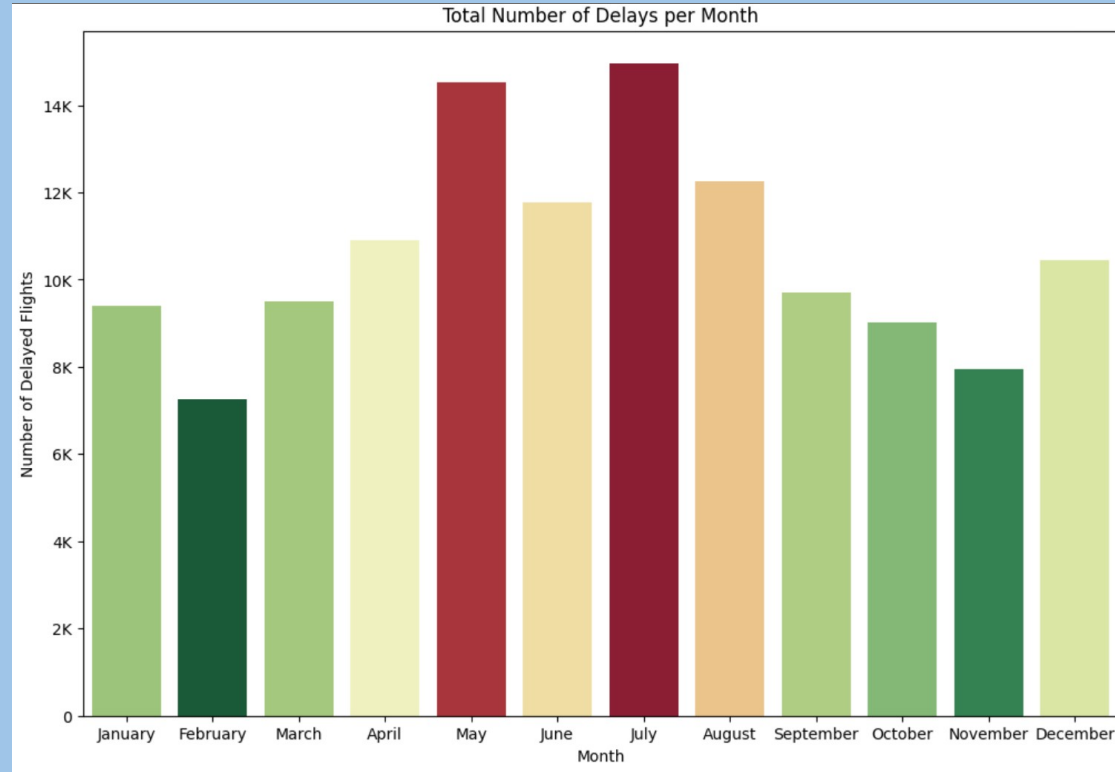
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Visual Exploration



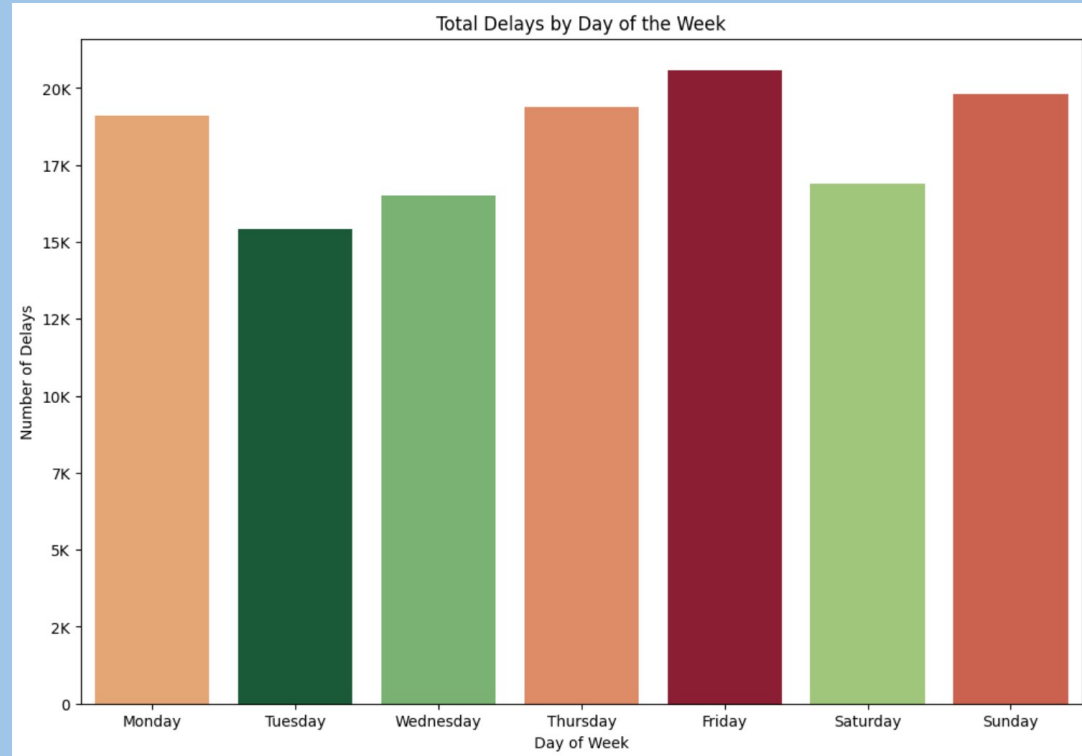
Delays by Month

- Delay frequency increased during the summer months
 - Peaked in July
- High travel demand means increased operational strain



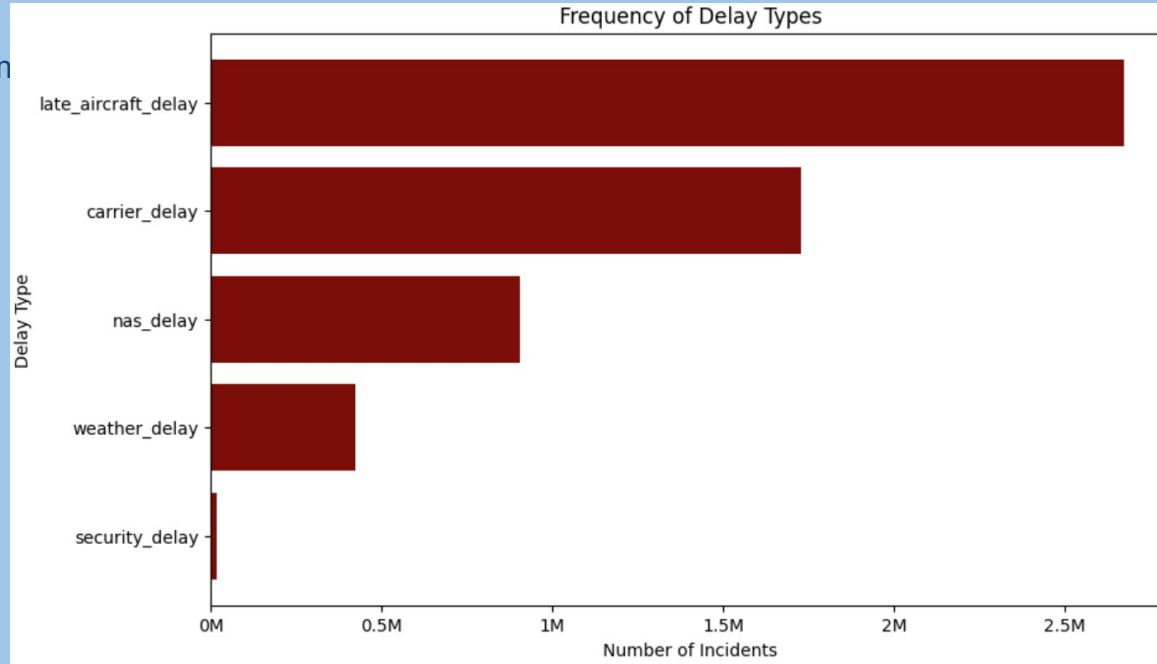
Delays by Day of Week

- Delays across the work week are largely the same
- Delays dip midweek and then experience a weekend uptick
 - Friday, Sunday highest
- Airports should prepare for extra traffic on peak days



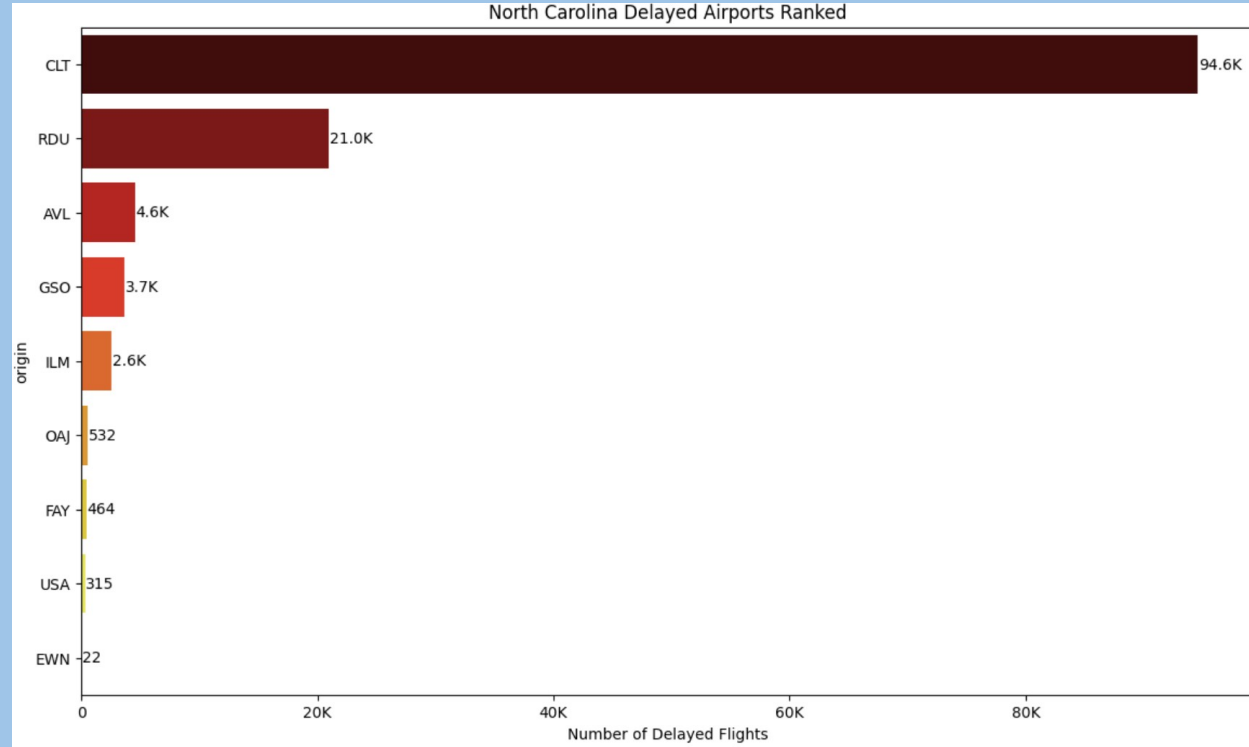
Frequency of Delay Types

- Late_aircraft often resulting from “ripple effects”
- Controllable factors → carrier and NAS delays are 2nd and 3rd in frequency
- Security and weather delays occur at a lower frequency



Airports Ranked

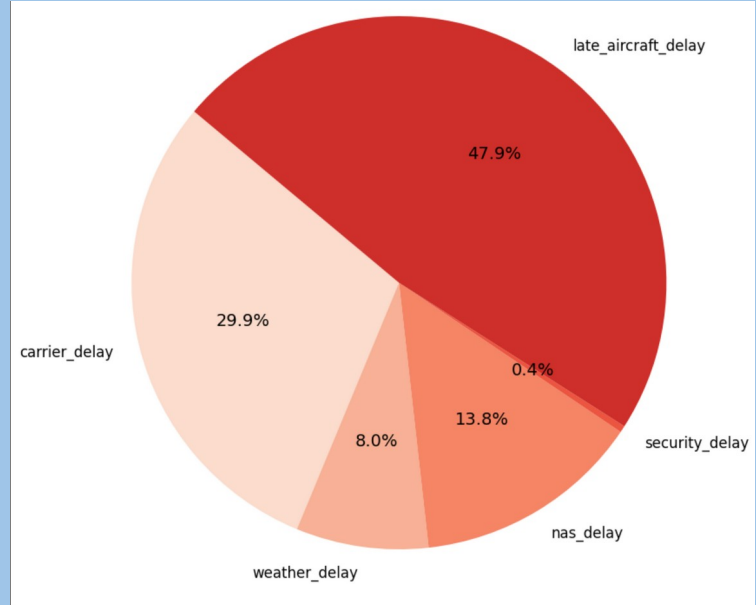
- CLT by far the leader in flights and delays
 - 94K incidents
- RDU ranks 2nd though total is lower than CLT's
- Prevention efforts should be focused on these two airports

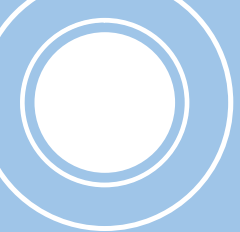


Top 5 Airports Analysis

Average Delay (Minutes) per Reason for Top 5 Airports

Airport	carrier_delay	weather_delay	nas_delay	security_delay	late_aircraft_delay
AVL	6.1	1.5	4.1	0.1	7.2
CLT	5.8	1.5	2.7	0.1	9.3
GSO	4.8	1.1	2.5	0.0	5.5
ILM	3.9	1.5	2.4	0.0	6.8
RDU	4.6	0.6	3.6	0.0	6.9





04

Modeling Strategy

Algorithm Selection

- Decision Tree
 - Simple and interpretable rule-based model
- Random Forest
 - Ensemble of many trees to reduce overfitting
- Logistic Regression
 - Baseline linear probability model
- XGBoost
 - Gradient boosting model optimized for predictive performance
 - used to sequentially correct errors to maximize predictive precision

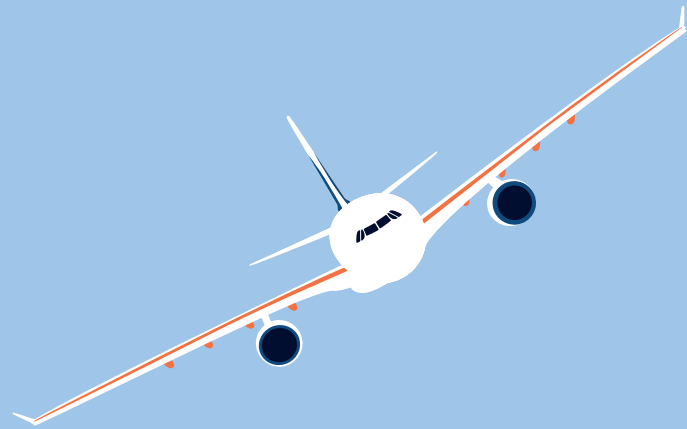


Model Fitting Process



- Partitioned data into 60% training and 40% validation sets
- Created custom target variable: `controllable_delay_binary`
 - Combined carrier, NAS, and security delays into controllable delays
 - Excluded weather and late aircraft delays from the target variable
- Delayed flights represented the minority class within the dataset
 - Tested balanced/scaled and unbalanced model configurations
 - Used balanced class weights to improve minority-class detection





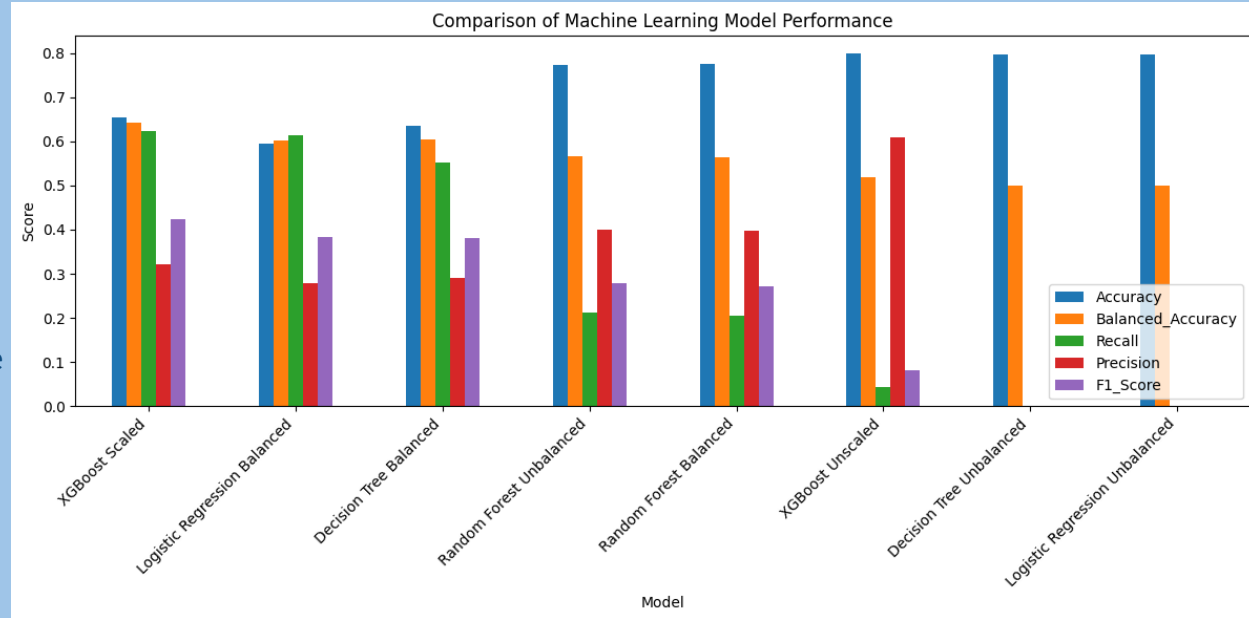
05

Evaluation & Results



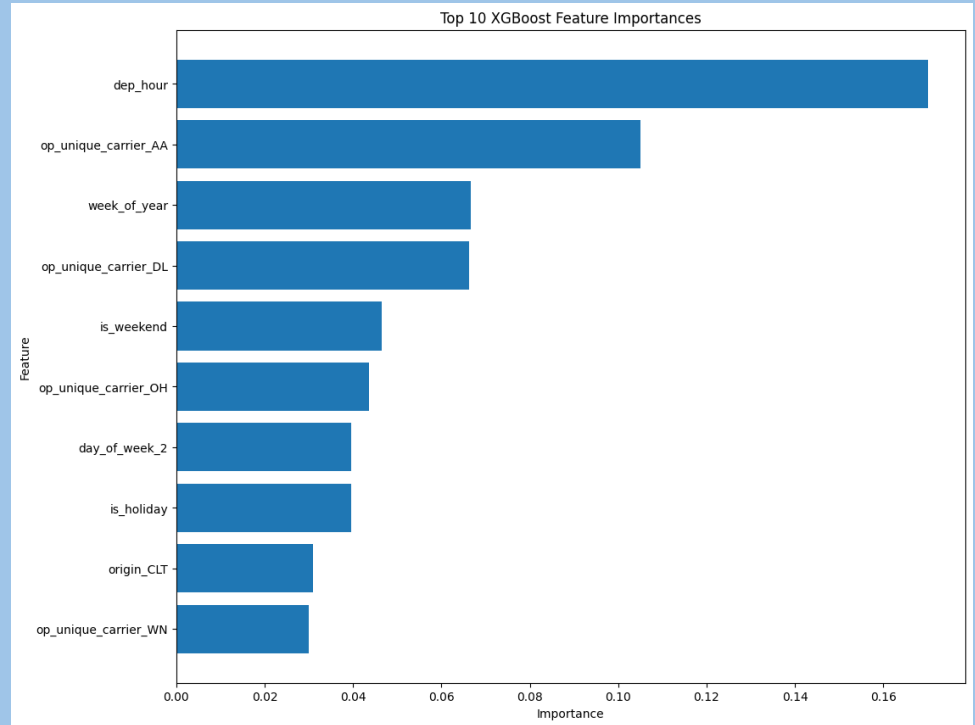
Performance Comparison

- Unbalanced models achieved high accuracy but performed poorly at identifying delayed flights (Naive Approach)
- Focused on recall, balanced accuracy, and F1 score rather than raw accuracy alone
- Balanced Random Forest was the second best model
- Scaled XGBoost was the best model
 - Achieved the strongest overall balance across metrics



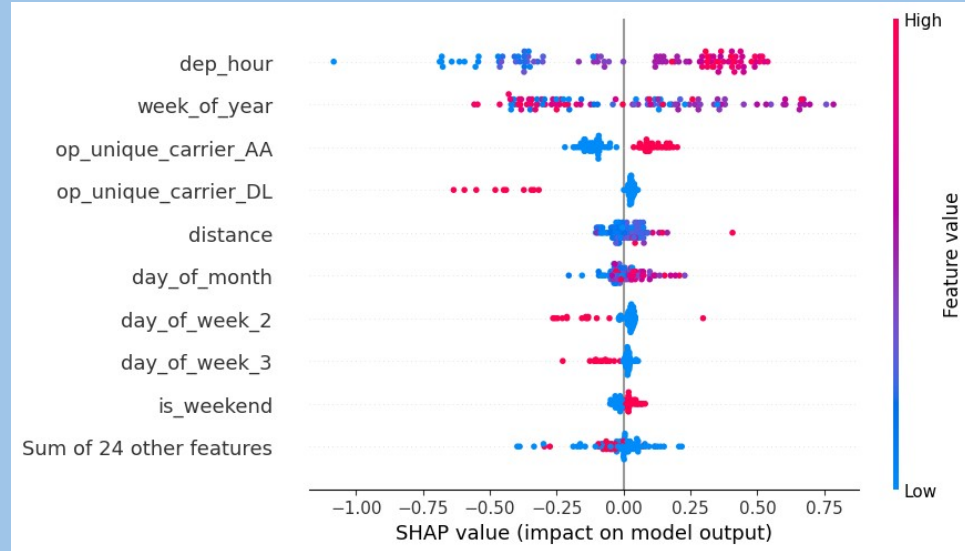
Global Predictors

- Departure hour was the most important predictor in the model
- Airline carrier also had a strong impact on delay prediction
- Week of year, weekends, and holidays influenced operational strain
- Charlotte (CLT) appeared as an important airport-related predictor
- Temporal and operational variables drove most predictions



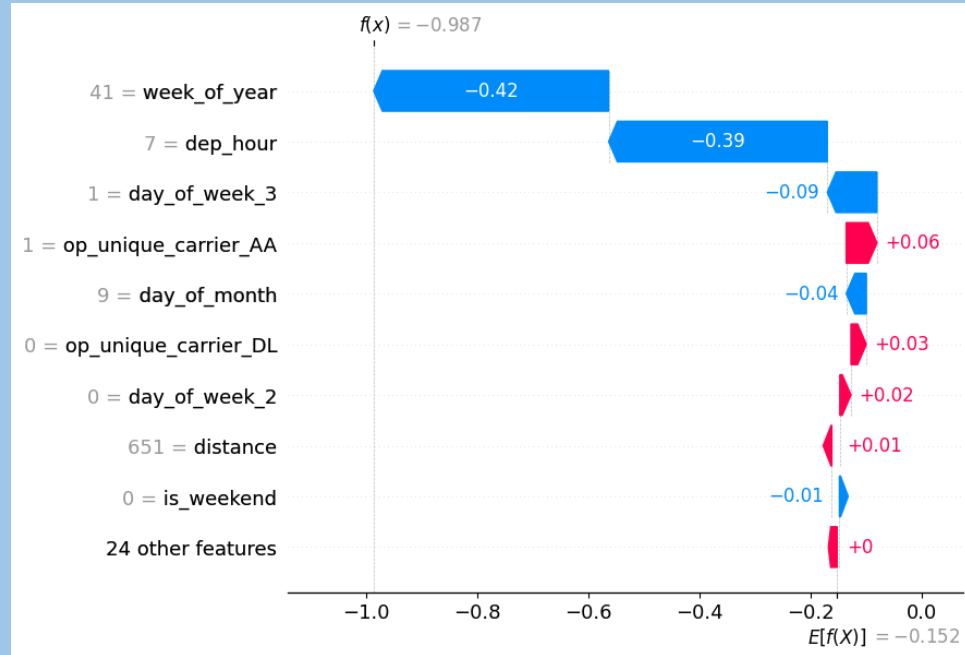
SHAP Beeswarm Plot

- Red points increased delay probability; blue points lowered it
- Later departure hours increased controllable delay risk
- Earlier departure hours generally reduced delay probability
- American Airlines flights often pushed predictions toward delays
- Operational timing variables consistently influenced predictions



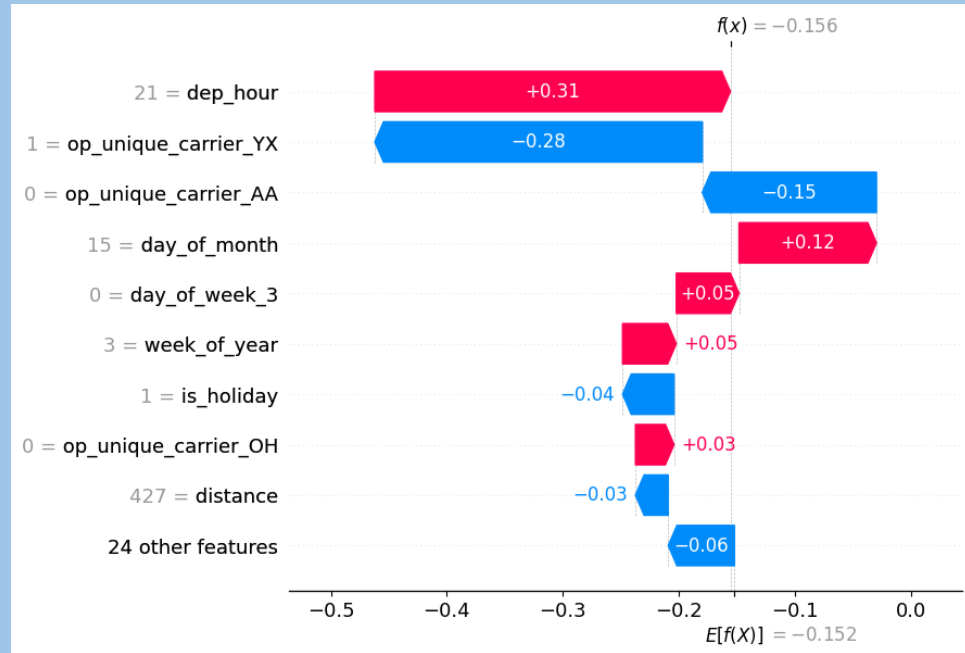
SHAP Waterfall Plot - Record 1

- Early morning departure strongly reduced predicted delay risk
- Week of year also lowered delay probability
- American Airlines slightly increased delay probability
- Overall prediction was strongly toward “no controllable delay”



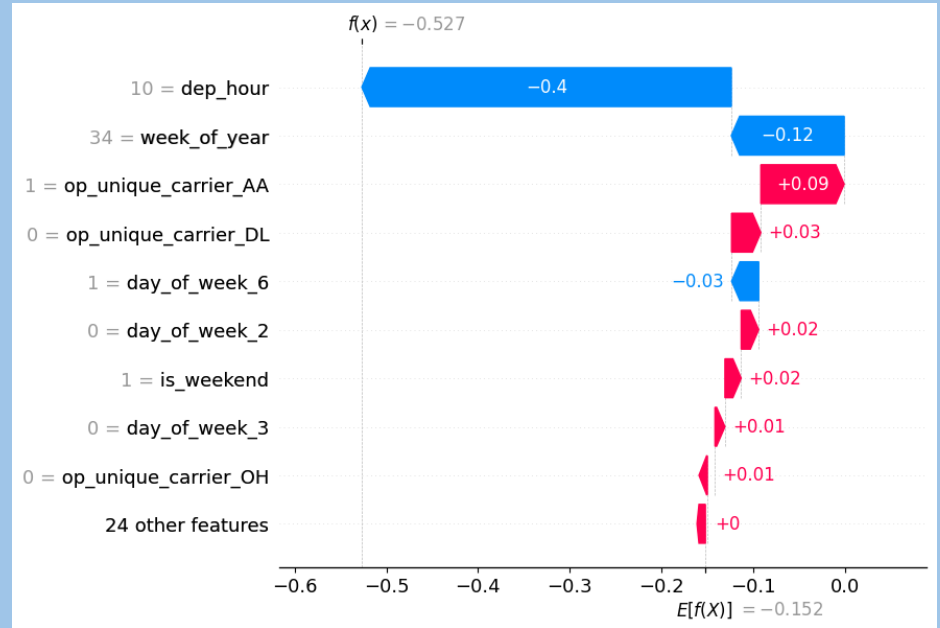
SHAP Waterfall Plot - Record 2

- Late departure hour increased delay probability
- Mid-month timing also pushed prediction toward delay
- Holiday variable slightly reduced delay probability
- Prediction moved closer toward a controllable delay outcome



SHAP Waterfall Plot - Record 3

- Departure hour again had the strongest impact on prediction
- Earlier departure time lowered delay probability
- American Airlines slightly increased delay probability
- Overall prediction still leaned toward no controllable delay



06

Conclusion & Recommendations



Summary Analysis

- Identifiable patterns
 - Temporal windows
 - Time of day
 - Day of the week
 - Airline carrier
 - Month
- System vulnerability
 - July as a peak summer travel month
- Targeted interventions
 - CLT departures on afternoon peak hours
 - Reallocation of resources



Recommendations

- Integration of a predictive model (XGBoost) into daily scheduling to create a proactive system
 - Adjust staffing, shifting focus to priority flights, adjust gate staffing
- Future data collection measures
 - Capture real-time metrics: aircraft turnaround times and ground crew staffing levels
 - Conduct targeted internal reviews into the specific late-afternoon time blocks and operational hubs
- Leveraging these models can improve passenger retention and reduce the direct financial costs of operation friction



Team Takeaways



- Group member ratings:
 - Maylee - Good
 - Katrina - Good
 - Owen - Good
- GenAI usage
 - Useful for adding value markers and specific color palettes on basic visualizations to enhance them
 - Useful for analyzing variables and how they might influence the dataset in data leakage and balancing data
 - Useful for understanding the links between probability thresholds and target metrics such as recall





Thank you!
Questions?